

House Price Prediction with Reliable Price Ranges and Clear Explanations

**PROJECT SYNOPSIS
OF MAJOR PROJECT**

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Introduction

Buying a house is the largest single purchase most families ever make, and almost nobody knows what the house they are looking at is actually worth. The usual way to find out is to ask an agent, look at what similar flats nearby sold for recently, and then argue about the difference. Banks face the same question when deciding how much to lend against a property. Because of this, a lot of effort has gone into building software that reads the details of a house, its size, number of bedrooms, age, condition, and locality, and estimates a price automatically.

These tools work reasonably well on average, and that average hides the problem. Zillow, the American property website, runs the best known of them and reports a typical error of about 2% for houses currently listed for sale but around 7% for houses that are not, where far less information is available. Seven per cent sounds small until it is applied to an actual property, where it becomes a gap of several lakh rupees. In 2021 Zillow used its own estimates to buy houses in bulk, then shut that business down after writing off 304 million dollars in a single quarter and cutting a quarter of its staff. Its chief executive said plainly that house prices had proved far harder to forecast than expected. The estimates were not useless. They were simply presented as though they were more certain than they were.

Most house price projects have the same shape. They train a model, report one error figure averaged over the whole test set, and stop. The person using such a system is handed a single number with nothing to judge it by: no sense of how wide the plausible range around it is, no explanation of which details pushed it up or down, and no way for the system to admit that a particular house is unusual enough that it should not be trusted at all. A confident-looking wrong number is worse than an honest “I am not sure about this one”, because the wrong number gets acted upon.

So this project builds a house price estimator that reports three things instead of one: a predicted price, a price range that has been tested to actually contain the true price about as often as it claims to, and a short explanation of which details of the house drove the estimate. On top of that, the system flags houses that are unlike anything it was trained on and tells the user to get a proper valuation instead of relying on the model. Predicting the price is the ordinary part. Being honest about how much to trust each individual prediction is the part we

think makes this worth doing.

Scope

This project works on residential property only, using two public datasets: the Ames Housing dataset from Iowa in the United States, and a public collection of listing prices from Indian metropolitan cities. The Ames data is used because it is well documented and widely studied, which makes it possible to check that our results are sensible. The Indian data is used because a price estimate for Iowa is of no practical use to anyone here. Commercial property, land without a building, and rental prices are out of scope.

The system estimates a price from a house's recorded characteristics at the time the data was collected. It does not forecast where prices are heading and does not track the market over time, a limit stated openly in the output: the model reflects the period its training data comes from, not today's market. It also does not scrape live listing websites, since their terms of use generally forbid it.

The deliverable is a demonstration web application together with an evaluation report, not a licensed valuation service, and every prediction carries a note saying so. Where the model performs badly, particularly on expensive or unusual properties, and where the Indian data proves too thin to support a reliable estimate, we plan to report that directly rather than leaving those cases out of the results.

Objectives

1. Clean and prepare both datasets, then build and compare a set of models, from plain linear regression as a baseline up to random forests and gradient boosting, measuring accuracy by cross-validation and reporting typical error in currency terms rather than only as an abstract score.
2. Attach a price range to every prediction using conformal prediction, then verify on held-out data that a range claiming to be correct 90% of the time really does contain the true price about 90% of the time. Report both the coverage achieved and the width, since a range that

is always correct but uselessly wide is no help.

3. Explain each prediction by naming the few features that pushed the estimate furthest up or down relative to a typical house, so the user sees the reasoning and not just the output.
4. Detect houses that do not resemble the training data, or whose range comes out unusually wide, and mark those predictions as low confidence with a recommendation to consult a professional valuer instead.
5. Build a small web application tying this together, and evaluate the system honestly, including separate results for cheap and expensive properties and for each city, so that a good overall average does not hide poor behaviour on one segment.

Literature Review

De Cock (Journal of Statistics Education, 2011) published the Ames Housing dataset, recording 2,930 property sales in Ames, Iowa between 2006 and 2010, described by a large set of variables covering the size, quality, age, and condition of each house. A split of it forms the well-known Kaggle house price competition, which uses 79 explanatory variables with roughly 1,460 houses for training and a similar number held back for scoring. This is our primary dataset, and its wide use means our numbers can be compared against published results rather than standing alone.

It was released as a replacement for the older Boston housing dataset, which is worth mentioning for a different reason: that dataset was removed from the scikit-learn library in version 1.2 on ethical grounds, because one of its variables was built on the assumption that racial segregation raised property prices. We avoid it entirely and take the lesson from it. Our system will not use caste, religion, or community composition as an input, even where a locality name might act as a proxy for one, and this limitation is noted explicitly.

Grinsztajn, Oyallon and Varoquaux (NeurIPS, 2022) benchmarked deep learning against tree-based methods across 45 tabular datasets and found that tree-based models such as gradient boosting and random forests remain the strongest choice on medium-sized tables of around ten thousand rows, before even accounting for their much lower training cost. Our datasets sit

squarely in that range. This is the reason the project deliberately does not use a neural network: on data of this shape and size, it would cost more and deliver less.

Rosen (Journal of Political Economy, 1974) set out the economic idea underlying price prediction of this kind: a house is a bundle of separate attributes, and its price is built up from what buyers will pay for each of them. This is why explaining a prediction feature by feature is meaningful rather than decorative.

Romano, Patterson and Candès (NeurIPS, 2019) and the survey by Angelopoulos and Bates (2021) describe conformal prediction, the main technique this project depends on. It turns any trained model into one that outputs a range instead of a single number, with a guarantee that the range contains the true value at the rate the user asks for. The guarantee assumes nothing about the shape of the data or the model, only that new houses resemble the ones used for calibration. The variant proposed by Romano and colleagues adapts the width of the range to each individual case, staying narrow for ordinary houses and widening where the model is genuinely uncertain, which is exactly the behaviour a buyer needs.

Lundberg and Lee (NIPS, 2017) introduced SHAP, which splits an individual prediction into the contribution of each input feature using a fair-division result from game theory. It works with gradient boosting models and produces per-house explanations rather than one global ranking of important features, so we use it for the explanation part of the system.

Finally, the closure of Zillow Offers in 2021, examined in the Journal of Information Systems Education (2024), is the clearest practical evidence for the problem addressed here: a well-built model with a respectable average error still caused large losses, because the business acted on individual estimates without accounting for how uncertain any one of them was.

Problem Statement

House price prediction systems, both commercial tools and the great majority of academic and student projects, output a single number and report a single average accuracy figure. A user receiving such an estimate has no way to tell whether this particular house is one the model handles well or one it is guessing at, no visibility into why it produced that figure, and no warning when the property is unusual enough to fall outside what the model has ever seen. As

far as we can determine, there is no simple, freely available system, and certainly none built for the Indian market, that returns a price together with a range whose reliability has been measured, a per-house explanation, and an explicit refusal to answer when it should not.

Feasibility Study

This is comfortably achievable in one semester by one student, mainly because the hard resource problem does not exist here. Both datasets are free, public, and small, a few thousand rows each, so the models train in seconds to minutes on an ordinary laptop processor. No GPU is needed at any stage. Every library involved is free and open source: scikit-learn for the groundwork, gradient boosting libraries for the main models, MAPIE for the price ranges, SHAP for the explanations, and Streamlit for the web interface. Free Google Colab is available as a fallback if the laptop proves slow, and free hosting tiers exist for putting the finished demo online.

The real risks lie in the data, not the computing. The Indian dataset is considerably messier than the Ames one: duplicate rows, amenity columns that use a filler value where nothing was recorded, and locality as free text with a long tail of names appearing once or twice. It also lists asking prices rather than confirmed sale prices and carries no transaction date, which is why the project claims only to model the period the data was collected in. The second risk is that the range guarantee holds only while new houses resemble the ones used to calibrate it. Rather than assume this, we test it, and where it breaks that becomes a reported finding rather than a hidden failure.

Methodology

The work proceeds in stages. First, the data is cleaned. Duplicate rows are removed and missing values are handled with attention to what they actually mean, since in the Ames data a blank often records that a house has no basement or garage rather than that the information is unknown. Filler codes in the Indian amenity columns become genuine missing values. Extreme outliers flagged in the original dataset documentation are removed, and both price and floor area are log-transformed, since both are heavily skewed and a few very expensive properties would otherwise dominate the fitting.

Second, features are prepared. Quality and condition ratings are mapped onto a numeric scale that preserves their order, locality is encoded in a way that handles rare names without leaking the target price into the inputs, and a few derived features are added such as total usable area and the age of the house at sale. Price per square foot is deliberately excluded, since it is computed from the answer and would produce impressive but meaningless accuracy.

Third, the data is split three ways rather than two: a training portion, a separate calibration portion, and a test portion. The calibration portion is used only to work out how wide the price ranges need to be, never to fit the model. Keeping these separate is what makes the range guarantee valid.

Fourth, the models are trained: linear regression as the baseline, then a random forest and a gradient boosting model, tuned by cross-validation on the training portion alone. Fifth, the calibration portion is used to build 90% price ranges, using both the basic conformal method and the adaptive quantile-based variant, so a fixed-width approach can be compared against one that widens for harder cases.

Sixth, explanations are generated with SHAP, giving each prediction a short list of the features that raised or lowered it against a typical house, plus a global summary of what the model relies on overall. Seventh, the confidence flag is assembled from three signals: the width of the range relative to the predicted price, how far the house sits from the training data, and whether important fields are missing. Crossing a threshold set on the calibration data marks the prediction low confidence.

Finally, the system is evaluated on the test portion, which is touched only once. We report typical error in currency terms, how often the 90% range actually contains the true price, and the average width of that range, comparing against the simpler alternatives: a model with no range at all, and one that puts a fixed percentage band around every prediction. Results are broken down by price bracket and by city, because we expect the model to be weaker on expensive properties and would rather show that than average it away. The interface is a simple web form returning the estimate, the range, the explanation, the confidence flag, and a clear statement that this is not a professional valuation.

Standards Adopted

The project is written in Python 3 and follows the scikit-learn conventions for models and data pipelines, which keeps every component interchangeable and testable. MAPIE, part of the scikit-learn ecosystem, provides the conformal prediction implementation, and SHAP provides the explanations, so the two central techniques come from peer-reviewed, maintained libraries rather than from code written from scratch for this project. Error is reported on the logarithm of price, matching the standard used by the Kaggle competition on this dataset, and also translated back into plain currency terms so that the numbers mean something to a reader who is not a statistician.

For reproducibility, random seeds are fixed, dependencies are pinned in a requirements file, and notebooks are numbered so they run in order from raw data to final results. Code is version-controlled with Git, functions carry docstrings, and a README explains setup so the guide and evaluators can run everything without needing to ask. Ethically, no feature based on caste, religion, or community composition is used, the problematic Boston housing dataset is avoided, and every output carries a disclaimer stating that it is a model estimate and not a professional valuation.

Facilities Required

A laptop with an internet connection is sufficient for the entire project. A free Kaggle account is needed to download the datasets, a free Google Colab account is kept available as a computing fallback, and a free GitHub account hosts the code. A free hosting tier such as Streamlit Community Cloud or Render may be used to put the demonstration application online. No paid software, no GPU, and no specialised hardware is required.

Expected Outcomes

By the end of the semester we expect a trained model whose accuracy on the Ames data is comparable to published results, and a first honest measurement of how well the same approach transfers to Indian listing data. On top of that, price ranges whose measured coverage lands

close to the 90% they promise, with widths small enough to be useful; per-house explanations naming the features that moved each estimate; and a confidence flag that catches a meaningful share of the cases where the model is worst, so that those users are sent to a valuer rather than given a bad number.

The tangible deliverables are a working web application, a reproducible code repository, and a project report that records the failures alongside the successes: where the ranges were too wide to be practical, where coverage fell short of what was promised, which kinds of property the model handled badly, and how much of the Indian data proved too sparse to model reliably. A system that knows the limits of its own estimates is the point of the project, and a report that hid its own limits would contradict that.

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